



SDA WEEK 3

USE CASES

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USE CASES

- In object-oriented analysis and design (OOAD), use cases are the primary technique for capturing functional requirements from the user's perspective.
- Key idea:
 - Use cases answer the question: What will the system do?

They do not describe how the system is implemented.

OBJECTIVES OF USE CASES

- Capture system functionality clearly
- Focus on user goals
- Provide a bridge between requirements and design
- Serve as input to class diagrams, sequence diagrams, and testing

USE CASE DEFINITION

A use case is:

- A scenario-based technique in UML
- A formal description of how a system interacts with external entities
- A collection of related success and failure scenarios
- A use case represents one complete user goal achieved through interaction with the system.

USE CASE DIAGRAM: GUIDELINES

- Use case names must start with a verb (e.g., Register User, not Registration)
- Avoid showing internal logic or sequence
- Do not show actor-to-actor communication
- Actors are named in singular form (Student, not Students)
- No personal names (John, Ali, etc.)

USE CASE DIAGRAM

Components of Use Case Diagram:

- Actors
- Use cases
- Relationships between them
- System boundary

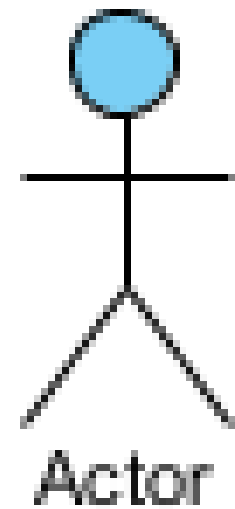
It provides a high-level functional view of the system.

ACTORS

Actors are external entities that interact with the system.

They can be:

- Human users
- External systems
- Hardware devices
- Timers or sensors

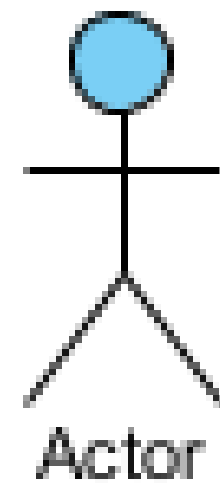


Notation

ACTORS

Types of Actors

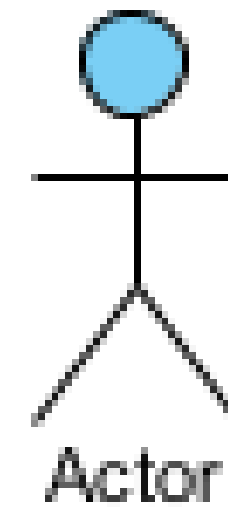
- Primary Actor: Initiates interaction to achieve a goal (e.g., Customer)
- Secondary Actor: Provides a service to the system (e.g., Payment Gateway)
- Offstage Actor: Has interest but no direct interaction (e.g., Tax Authority)



ACTORS

Key questions to identify actors:

- Who uses the system?
- Who supplies data?
- Who receives results?
- Who maintains the system?



USE CASE

A use case:

- Represents a complete system functionality
- Starts when an actor initiates it
- Ends when the goal is achieved or fails

Examples:

- Login User
- Transfer Funds
- Generate Report

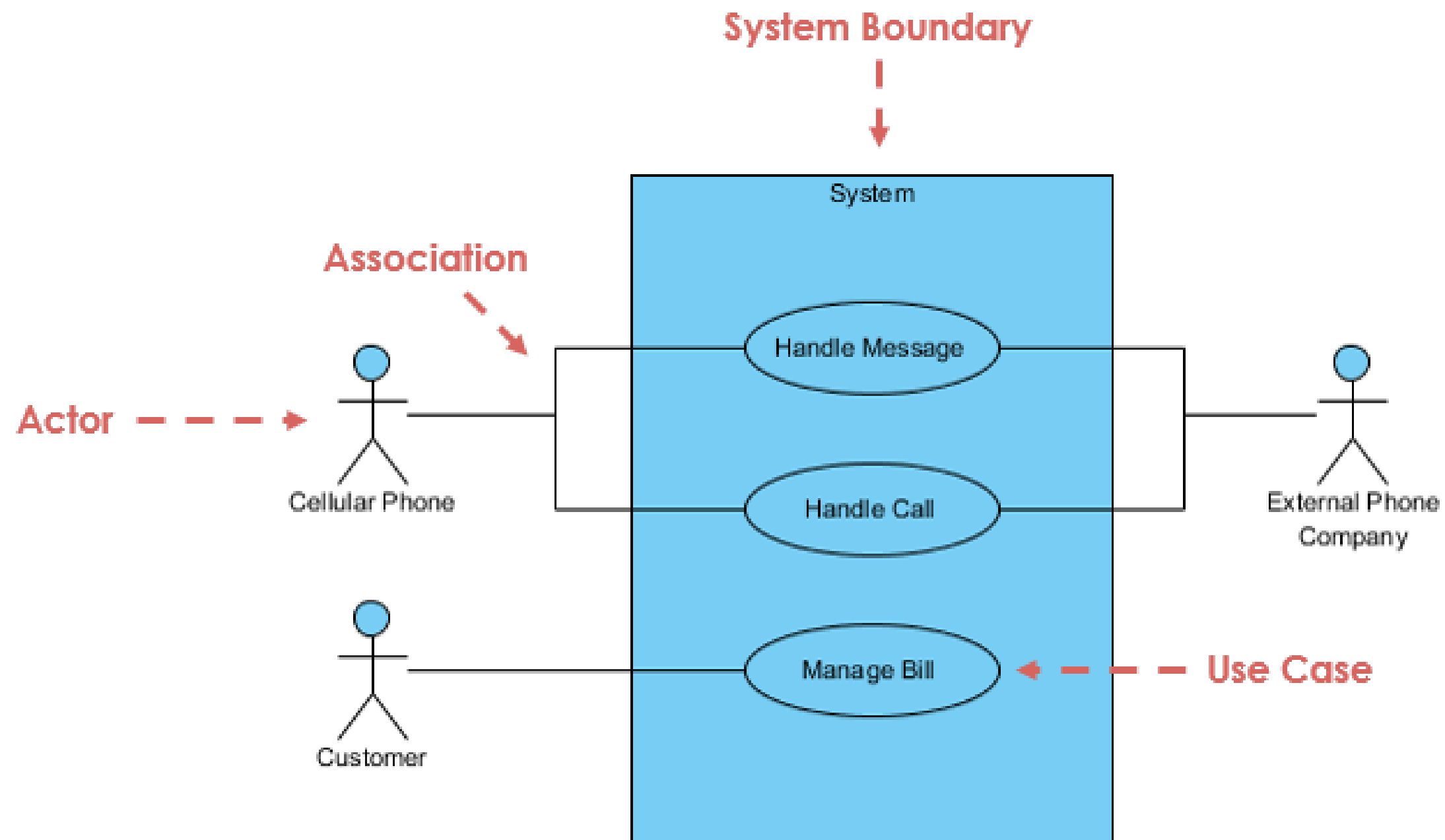


Notation

COMMUNICATION LINK

- The participation of an actor in a use case is shown by connecting an actor to a use case by a solid link.
 - Actors may be connected to use cases by associations, indicating that the actor and the use case communicate with one another using messages.
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USE CASE DIAGRAM AT A GLANCE

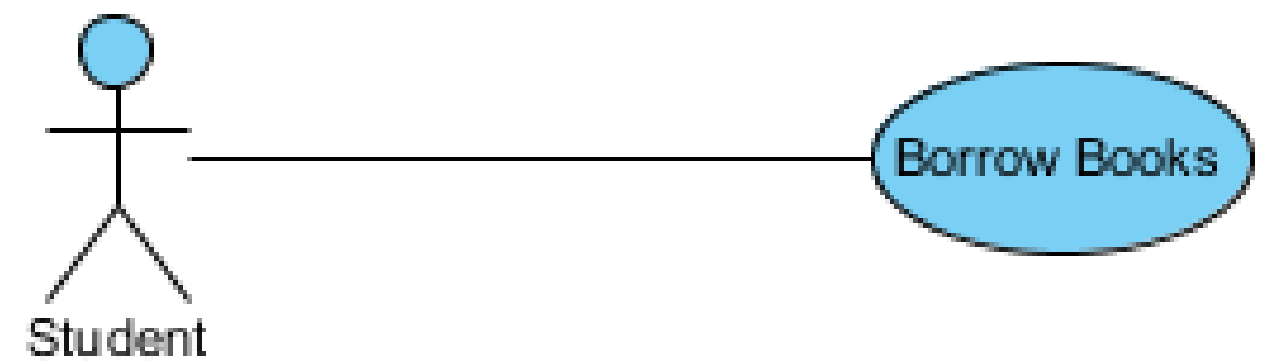


RELATIONSHIPS IN USE CASE DIAGRAMS

- **Association** between an actor and a use case
- **Generalization** of an actor
- **Generalization** of a use case
- **Extend** relationship between two use cases
- **Include** relationship between two use cases

ASSOCIATION RELATION

- Simple communication link
- Represented by a solid line
- Example:
 - Student – Borrows books
 - Customer - withdraws Cash



GENERALIZATION RELATION

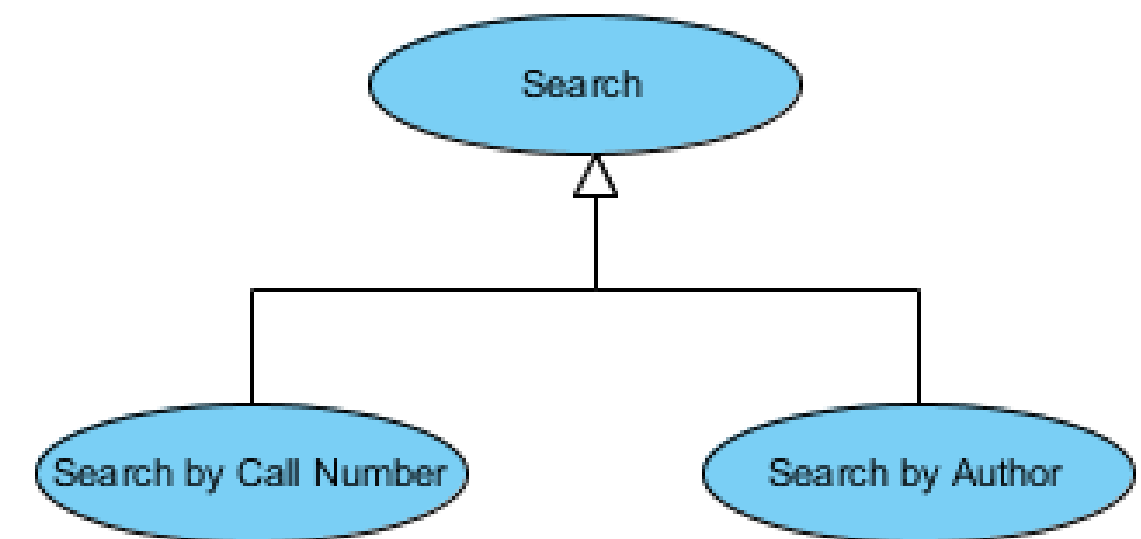
1. Actor Generalization

- One actor inherits the behavior of another
- Used when actors share common roles
- Example:
 - User
 - Student
 - Admin

GENERALIZATION RELATION

2. Use Case Generalization

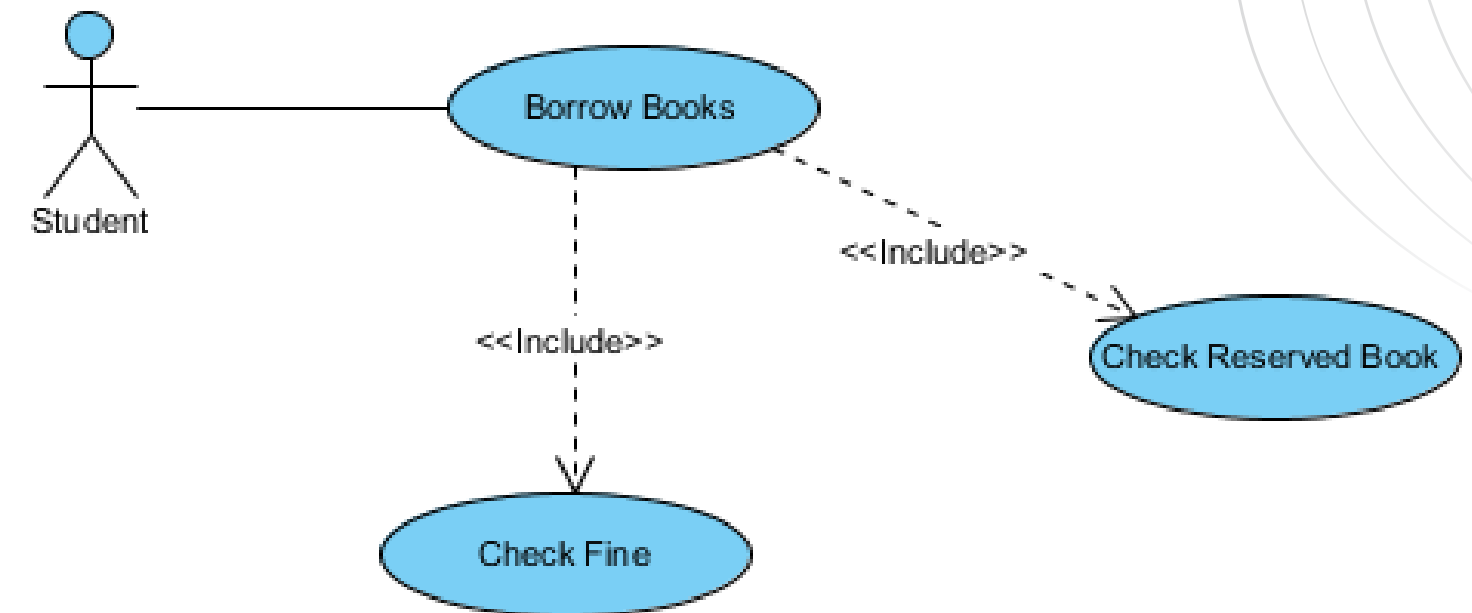
- Child use case inherits behavior of parent
- Used when use cases share common functionality
- Example:
 - Make Payment
 - Pay by Cash
 - Pay by Card



INCLUDE RELATION

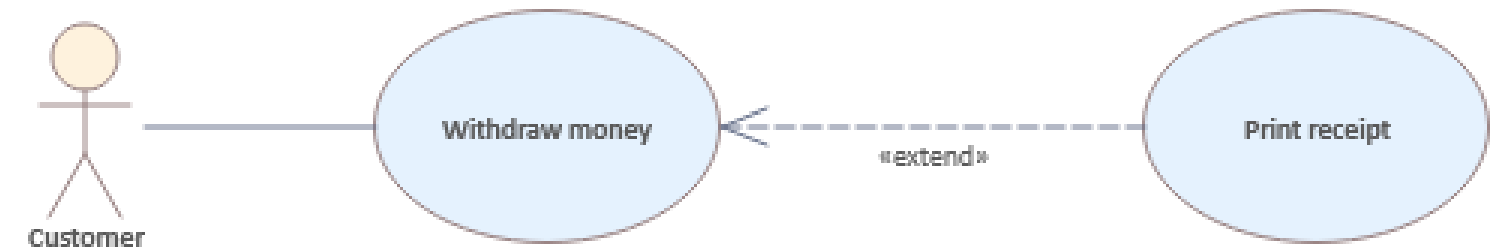
<<include>> relationship represents behavior that is factored out of a use case.

- Purpose
 - Extract common mandatory behavior
 - Promote reuse
- Characteristics
 - Always executed
 - Base use case depends on included use case
- Use when:
 - The behavior is required every time
- Example:
 - Borrow Book ---<<include>>--- Check Fine

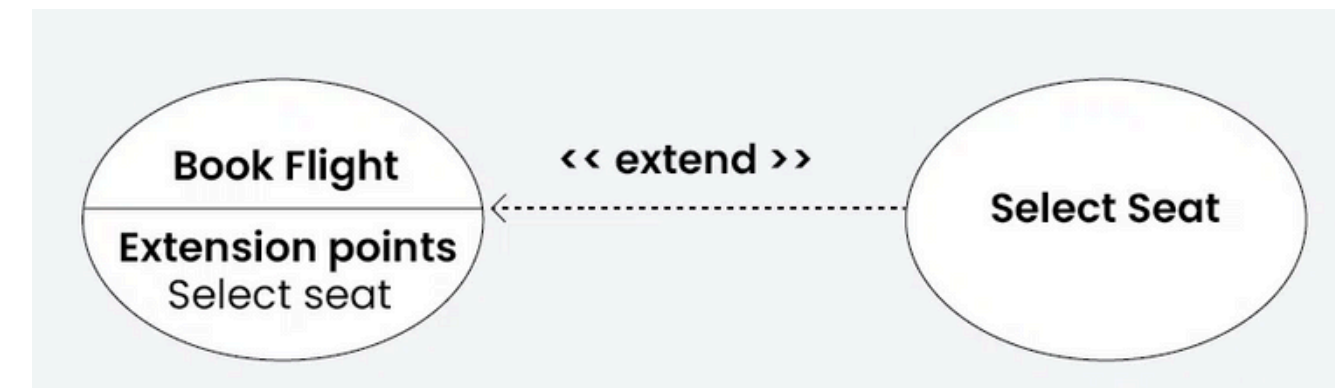


EXTEND RELATION

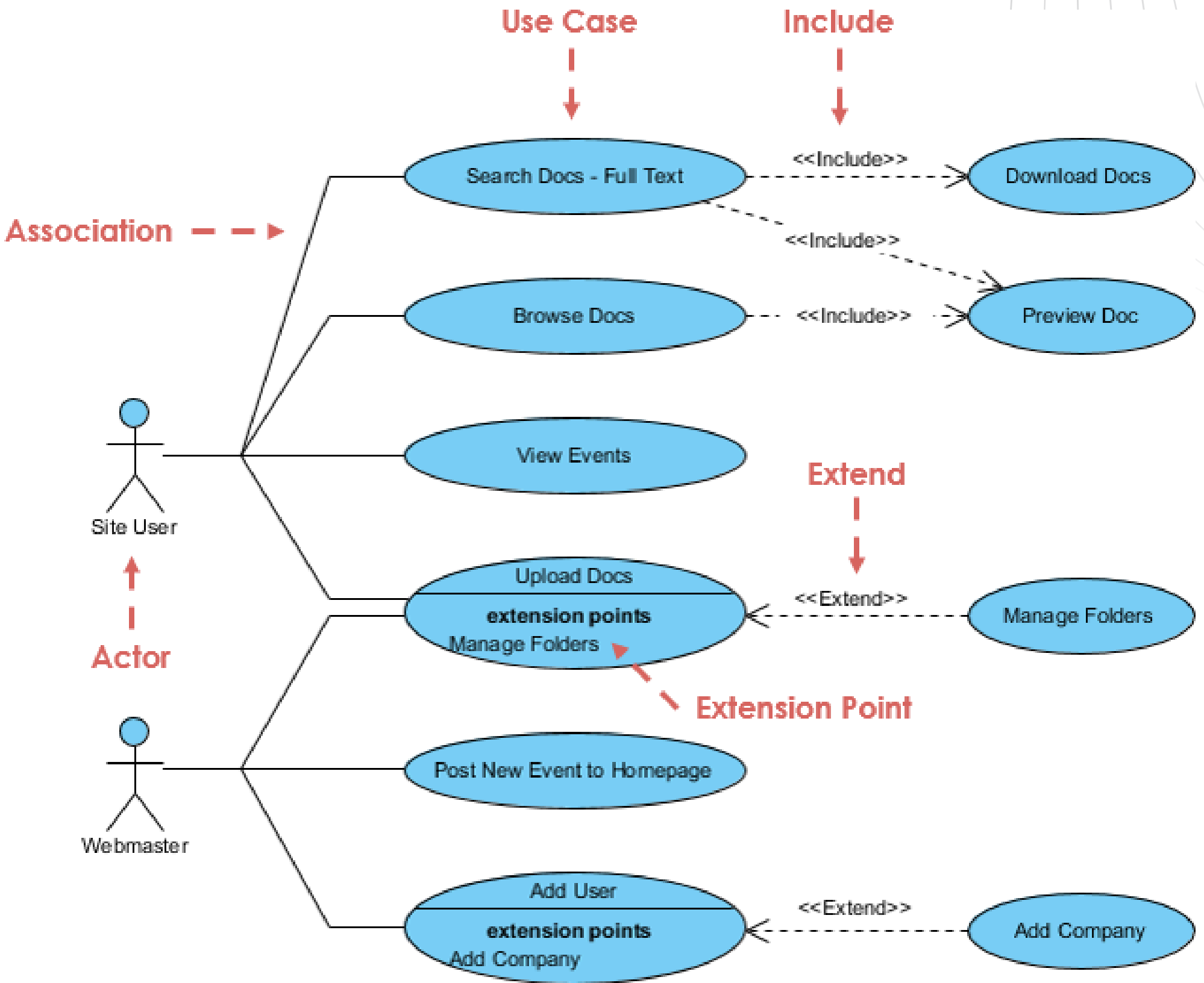
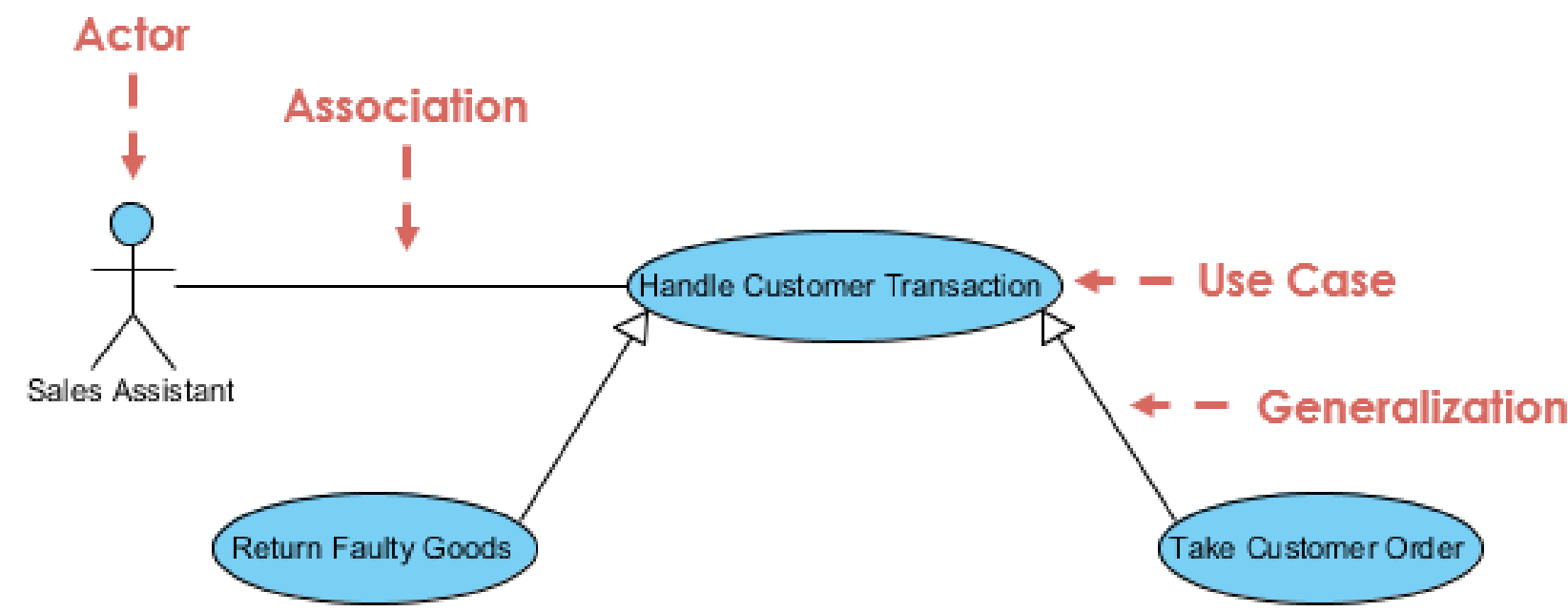
<<extend>> relationship represents exceptional or seldom invoked cases.



- Purpose
 - Model optional or conditional behavior
- Characteristics
 - Executed only under certain conditions
 - Extending use case depends on base use case
- Use when:
 - Additional behavior may or may not occur
- Example:
 - Withdraw Cash ---<<extend>>--- Print receipt



EXAMPLES



EXERCISE

Automated Teller Machine (ATM) is a banking system. This banking system allows customers or users to have access to financial transactions. These transactions can be done in public space without any need for a clerk, cashier, or bank teller. The user is authenticated when enters the plastic ATM card in a Bank ATM, Then enters the user name and PIN (Personal Identification Number). For every ATM transaction, a Customer Authentication is required and essential. User checks the bank balance as well as also demands the mini statement about the bank balance if they want. Then the user withdraws the money as per their need. If they want to deposit some money, they can do it. After complete action, the user closes the session. If there is any error or repair needed in Bank ATM, it is done by an ATM technician. ATM technician is responsible for the maintenance of the Bank ATM, upgrades for hardware, firmware or software, and on-site diagnosis.

IDENTIFY ACTORS

Primary Actor

- Customer
- → Initiates transactions and achieves goals (withdraw, deposit, check balance)

Secondary / Supporting Actor

- Bank System / Authentication System
- → Verifies card, PIN, and account details

Technician Actor

- ATM Technician
- → Performs maintenance, upgrades, and repairs

IDENTIFY USE CASES

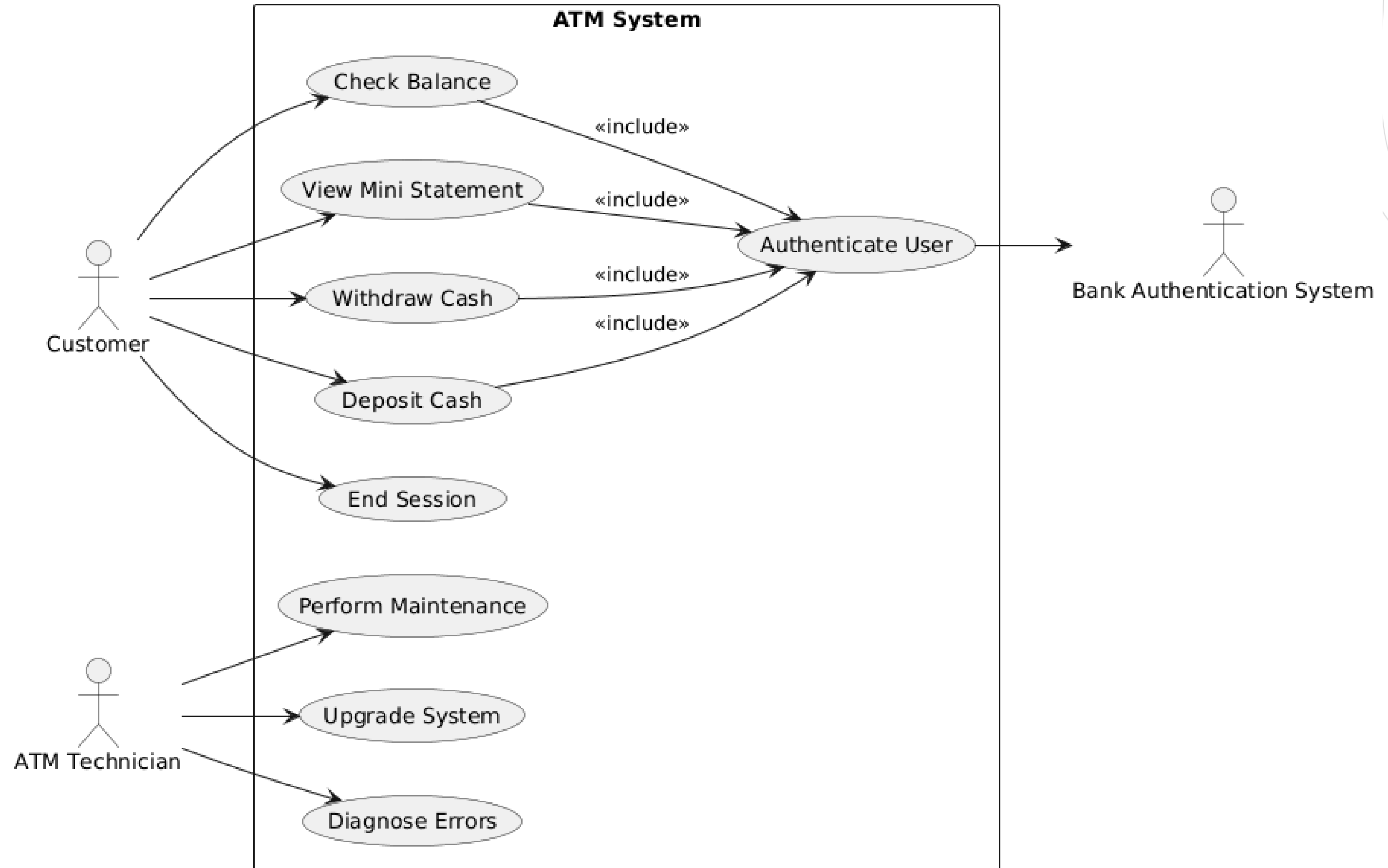
Customer Use Cases

- Authenticate User
- Check Balance
- View Mini Statement
- Withdraw Cash
- Deposit Cash
- End Session

Technician Use Cases

- Perform Maintenance
- Upgrade ATM Software/Firmware
- Diagnose ATM Errors

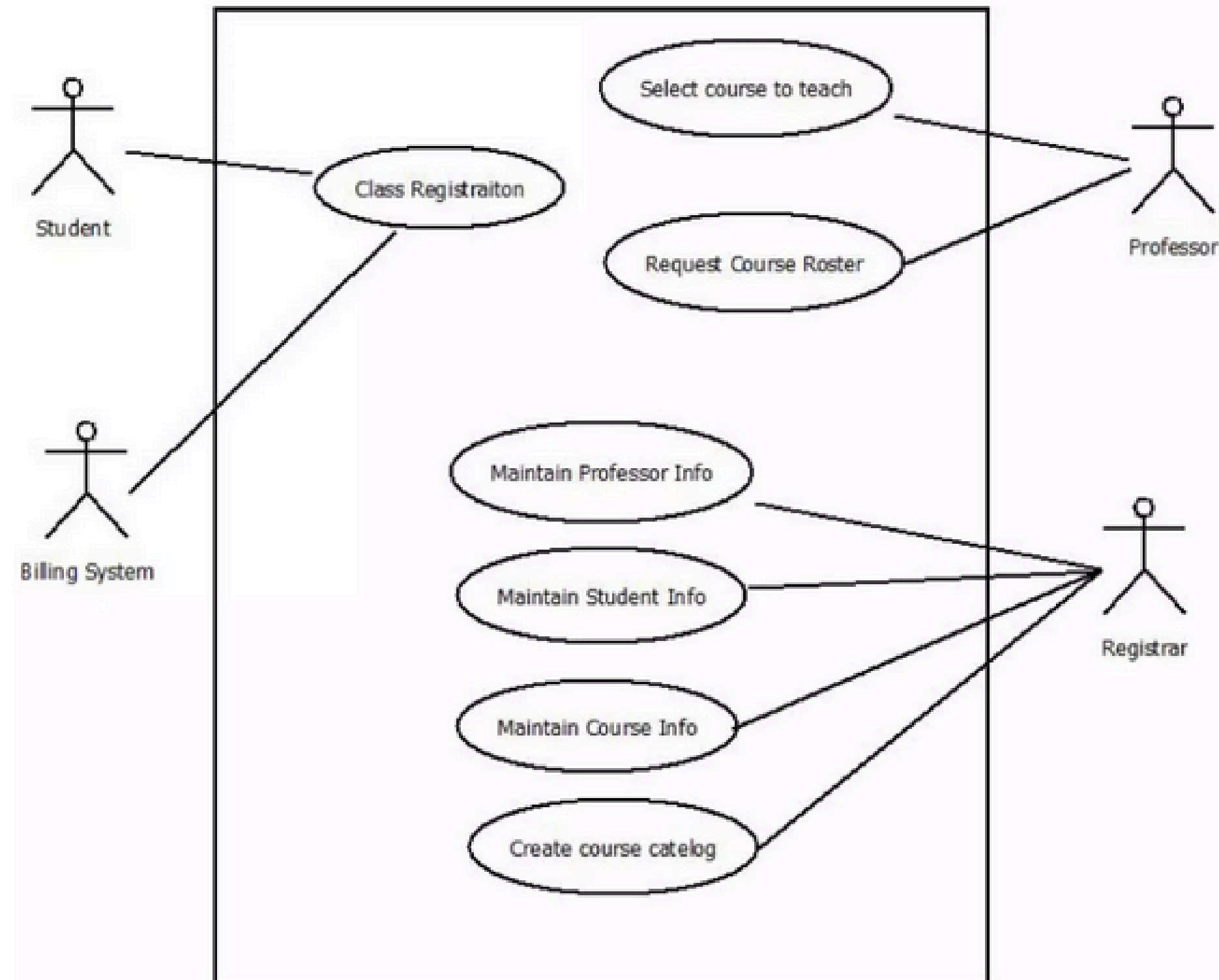
SOLUTION



EXERCISE 2

Create a use Case Diagram for an Online University Registration System. Professors indicate which courses they will teach on-line. A course catalog is created by the Registrar and can be printed. Registrar is also responsible to maintain the information about the professor, students, and course. Students are allowed to select online four courses for upcoming semester. No course may have more than 10 students or less than 3 students. When the registration is completed, the system sends information to the billing system. In addition, Professors can request for course rosters (students registered for the course) online. Students can add or drop classes online during the semester

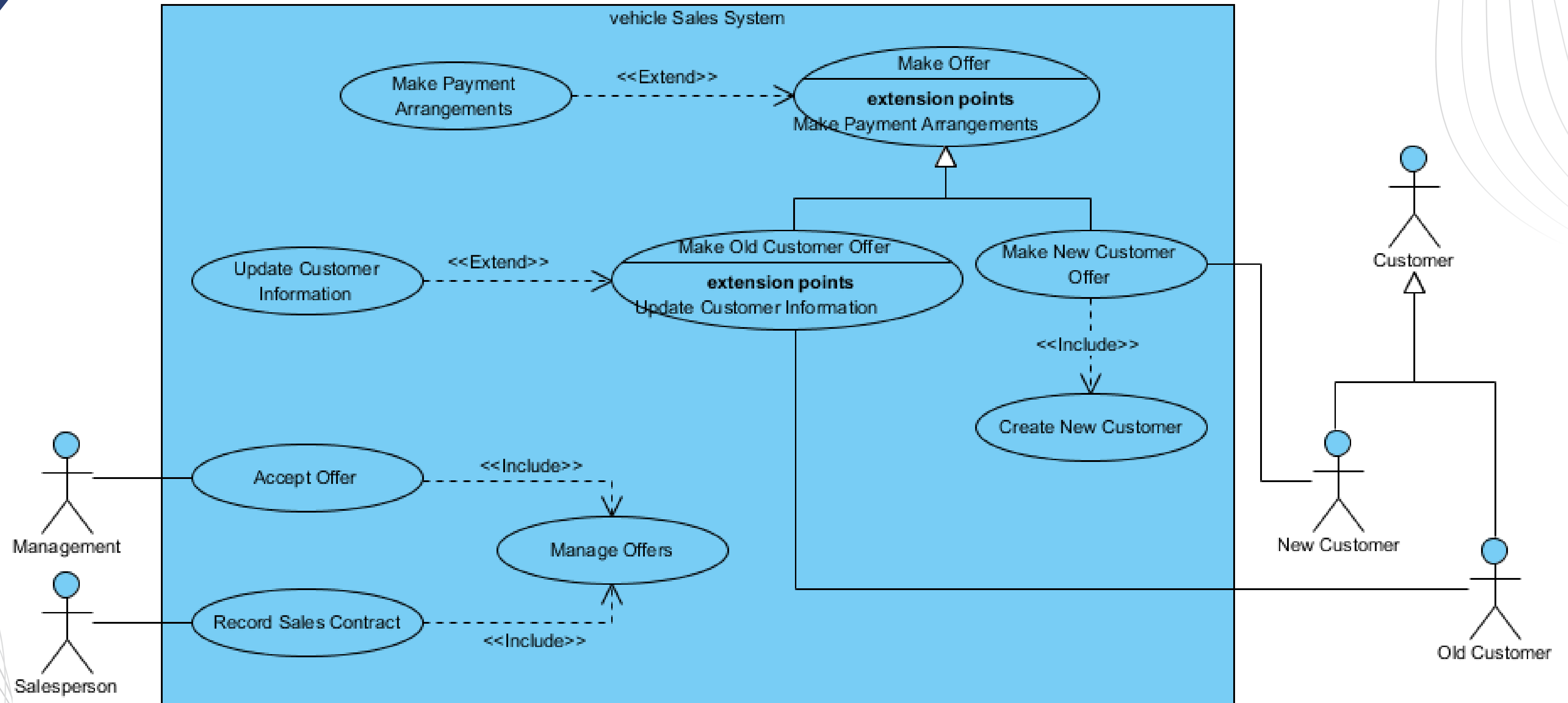
EXERCISE 2



EXERCISE 3

Draw a use case diagram for the **vehicle sales system**. Customer makes offer for the vehicle. Customer can be new customer or old customer. New and old customer can make their own offers. For every customer they have to get registered. System can update the existing customer information as well. Customer make payment if his/ her offer is accepted. Management has right to accept or reject the offer by managing the offer. Sales person records the sales contract of the accepted offer.

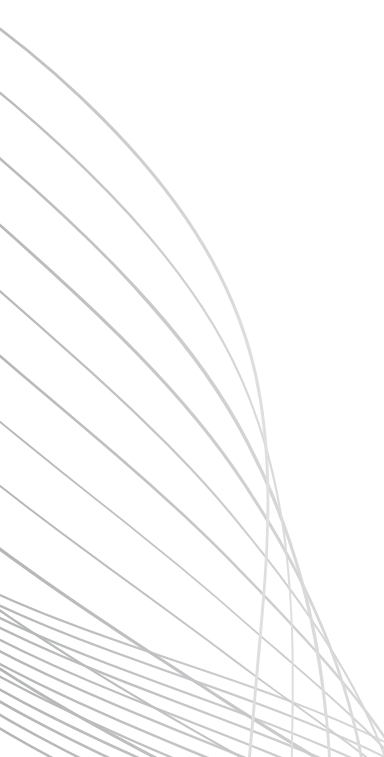
EXERCISE 3



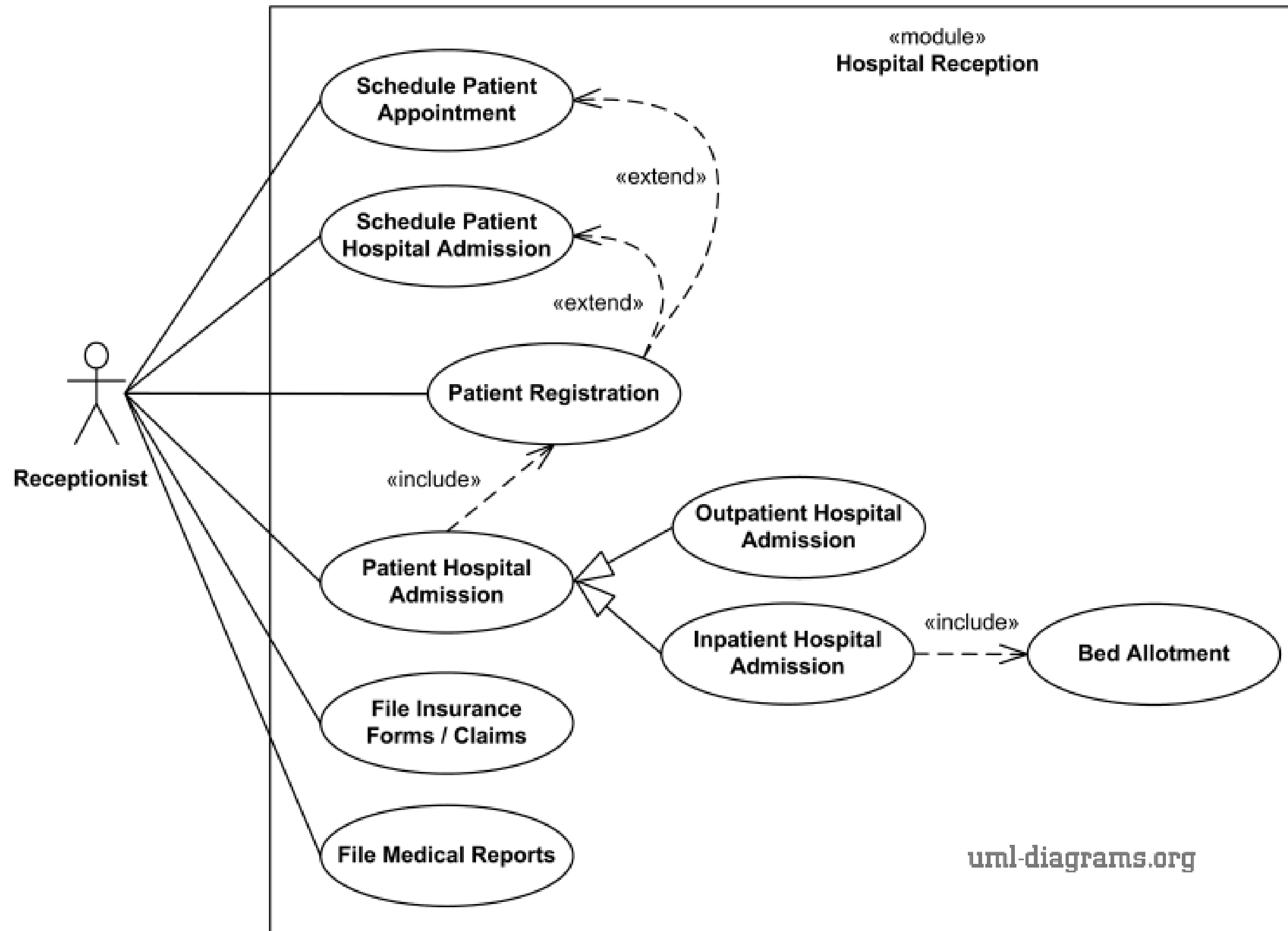


EXERCISE

Hospital Reception subsystem or module supports some of the many job duties of hospital receptionist. Receptionist schedules patient's appointments and admission to the hospital, collects information from patient upon patient's arrival and/or by phone. For the patient that will stay in the hospital ("inpatient") she or he should have a bed allotted in a ward. Receptionists might also receive patient's payments, record them in a database and provide receipts, file insurance claims and medical reports.



EXERCISE



SCENARIO

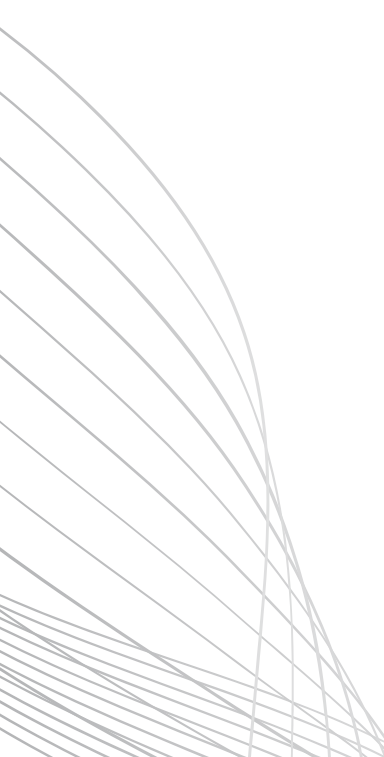
Ayesha is a first-semester undergraduate student studying Software Engineering at a large university. Most of her core courses are conducted in the engineering block, which she is familiar with. However, one of her elective courses, “Introduction to Psychology”, is conducted in a different building on campus that she has never visited before.

Ayesha opens the University Mobile App to check her course schedule. She views the course details, including lecture timings and the classroom location displayed on a campus map. While attending the course, she reads reviews and comments posted by other students enrolled in the same course. She finds a helpful comment by Amna saying, “Slides are very helpful for exams.”

Ayesha visits Amna’s profile and sends her a friend request. Amna receives a notification and accepts the request. Ayesha is notified about the acceptance and then explores the list of courses Amna is currently taking. She finds another course, “Cognitive Science”, which interests her and saves it to her favorites list for future reference.



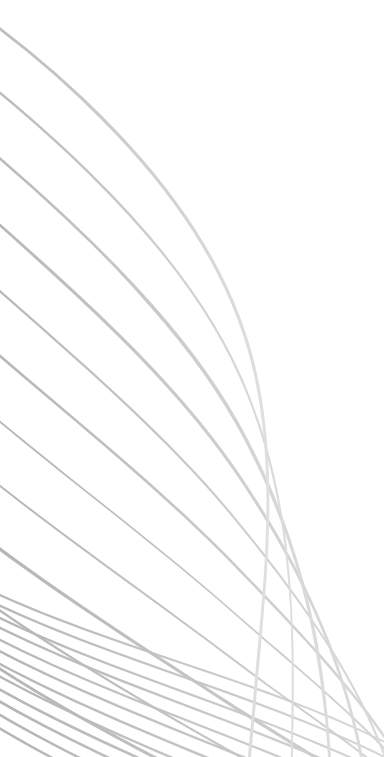
TASKS

1. Identify the actors involved in the system
 2. Identify the functional requirements of the system
 3. Write the functional requirements in clear “system shall” statements
- 



SOLUTION

Actors are roles interacting with the system, not people.

- Student (Primary Actor)
 - Notification Service (Secondary Actor - implicit)
 - University System / Database (Secondary - implicit)
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SOLUTION

Functional Requirements

- 1.The system shall allow students to view their enrolled courses.
- 2.The system shall display course details, including lecture time and classroom location.
- 3.The system shall show classroom location on a campus map.
- 4.The system shall allow students to view comments and reviews for a course.
- 5.The system shall allow students to view other students' profiles.
- 6.The system shall allow students to send friend requests.
- 7.The system shall notify a student when a friend request is received.
- 8.The system shall allow a student to accept or reject friend requests.
- 9.The system shall notify a student when their friend request is accepted.
- 10.The system shall allow students to view courses taken by their friends.
- 11.The system shall allow students to save courses to a favorites list.

USE CASE FORMATS AND TEMPLATES

Use Case Formats

- **Brief:** One-paragraph summary.
- **Casual:** Informal, multiple paragraphs.
- **Fully Dressed:** Detailed, structured, with pre/post conditions, flows, exceptions.

BRIEF USE CASE

Template

- Use Case Name
- Summary Paragraph

BRIEF USE CASE

Example 1: ATM - Withdraw Cash

Withdraw Cash:

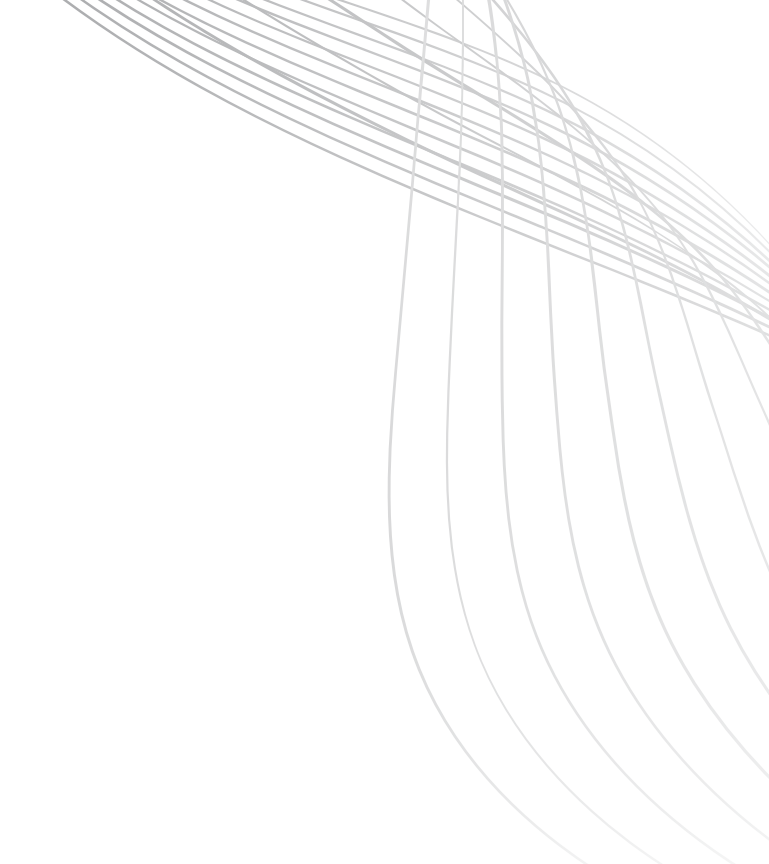
A customer inserts an ATM card and enters a valid PIN. The system authenticates the customer, allows the customer to enter a withdrawal amount, dispenses the requested cash, updates the account balance, and ends the session.

BRIEF USE CASE

Example 2: Online Shopping - Place Order

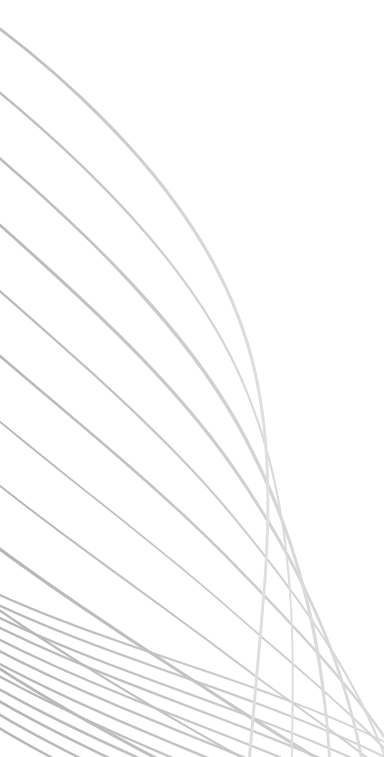
Place Order:

A customer selects items from the online store, reviews the cart, provides delivery and payment details, confirms the order, and receives an order confirmation from the system.



BRIEF USE CASE

When to Use Brief Use Cases

- Initial project scoping
 - Brainstorming system functionality
 - When time is limited
- 

CASUAL USE CASE

Template (Casual Use Case)

- Use Case Name
- Main Success Scenario (Paragraph)
- Alternate Scenarios (Paragraphs)

CASUAL USE CASE

Example 1: Library System - Borrow Item

Borrow Item:

A library member requests to borrow an item from the library catalog. The system verifies the member's identity and checks whether the borrowing limit has been reached. If the item is available, the system records the checkout date and assigns a due date. The member receives confirmation of the successful borrowing.

If the member has already borrowed the maximum allowed number of items, the system informs the member that borrowing is not permitted. If the item is not available, the system displays an appropriate message.

CASUAL USE CASE

Example 2: University System - Course Registration

Register for Course:

A student browses the course catalog and selects a course for registration. The system checks prerequisites and seat availability. If the conditions are satisfied, the system registers the student and updates the course roster. If prerequisites are not met or seats are full, the system informs the student and does not complete the registration.

CASUAL USE CASE

When to Use Casual Use Cases

- Requirements workshops
- Discussions with stakeholders
- Medium-level documentation

FULLY DRESSED USE CASE

Definition

A fully dressed use case is a complete, structured description of system behavior, including conditions, flows, and exceptions.

Characteristics

- Formal structure
- Step-by-step flow
- Includes preconditions and postconditions
- Used for design, testing, and validation

FULLY DRESSED USE CASE

Template (Fully Dressed Use Case)

- Use Case Name
- Primary Actor
- Stakeholders & Interests (optional)
- Preconditions
- Postconditions
- Main Success Scenario
- Alternate Flows
- Exception Flows

EXAMPLE 1

Field	Description
Use Case Name	Withdraw Cash
Primary Actor	Customer
Secondary Actor	Bank Authentication System
Brief Description	Customer withdraws cash from the ATM after successful authentication.
Preconditions	ATM is operational; Customer has a valid ATM card
Postconditions	Cash is dispensed; Account balance is updated; Session is terminated
Main Success Scenario	1. Customer inserts ATM card 2. System prompts for PIN 3. Customer enters PIN 4. System authenticates customer 5. Customer enters withdrawal amount 6. System dispenses cash 7. System updates account balance 8. System ends session
Alternate Flow	4a. Invalid PIN entered → System displays error and requests PIN again
Exception Flow	5a. Insufficient balance → Transaction cancelled and message displayed

EXAMPLE 2

Field	Description
Use Case Name	Submit Assignment
Primary Actor	Student
Secondary Actor	Learning Management System
Brief Description	Student submits an assignment electronically before the deadline.
Preconditions	Student is logged in; Assignment submission window is open
Postconditions	Assignment is stored; Submission confirmation is generated
Main Success Scenario	1. Student selects assignment 2. Student uploads solution file 3. System validates file format 4. System stores the submission 5. System confirms successful submission
Alternate Flow	3a. Invalid file format → System requests re-upload
Exception Flow	4a. System error occurs → Submission fails and error message is shown

EXAMPLE 2

Field	Description
Use Case Name	Withdraw Cash
Primary Actor	Customer
Secondary Actor	Bank Authentication System
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THANK YOU